

SoK: The Landscape of Post-Quantum Multi-Party Signature Aggregation

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Abstract. Post-quantum signature schemes such as ML-DSA, FN-DSA (Falcon), and SLH-DSA have significantly larger keys and signatures than their classical counterparts, making compact multiparty signing an increasingly important research problem. We survey more than 30 schemes published between 2021 and 2026, covering threshold and distributed signing, proof-based aggregation, and algebraic half-aggregation. Our survey shows that most existing work is designed around blockchain and validator-consensus applications, where large numbers of mostly homogeneous signers sign a common message and accountability is typically represented by a compact signer set.

We found that a different and practically important setting has received little attention: *institutional multiparty document signing*, where tens to hundreds of signers may use different post-quantum schemes or parameter sets during a gradual migration, while requiring non-interactive signing and exact per-signer accountability. To study this gap, we develop an eight-dimensional taxonomy based on deployment requirements and systematically classify existing approaches. We also formulate three testable hypotheses and propose a common benchmark methodology for evaluating communication, computation, verification cost, scalability, signer heterogeneity, and accountability. Our analysis finds no published study that quantitatively evaluates the institutional workload across the aggregation paradigms covered in our survey. We therefore identify the key experimental and architectural requirements for assessing whether existing post-quantum aggregation techniques can support heterogeneous, accountable, and non-interactive institutional signing.

Keywords: Post-quantum cryptography, threshold signatures, aggregate signatures, systematization of knowledge, ML-DSA, lattice-based cryptography

1 Introduction

Recent work on post-quantum signature compression has largely focused on a target workload, and it is worth naming that workload before diving into the literature it produced, because the workload is doing more explanatory work than any single technical choice inside these papers. Despite this diversity, existing

work is strongly concentrated on one workload: *large-scale blockchain and validator consensus*. Threshold ML-DSA constructions [3,13,22,21,24] target validator-set key management, while lattice-based aggregation of Falcon signatures [1,4] has been evaluated at scales of up to 2^{17} signatures. Hash-based STARK aggregation [23,15] has likewise developed primarily around blockchain applications, including an associated Ethereum Improvement Proposal [12]. These works address a genuine scalability challenge, but they leave open a fundamental question: *do techniques designed for large, homogeneous validator populations also address the requirements of institutional multi-party signing?*

Institutional document signing presents a materially different setting. A bank may require signatures from three internal authorities before a loan document becomes binding. A government workflow may require approval from several ministries, each operating its own certificate infrastructure. A multi-party contract may require every participating legal entity to sign, with the signatures remaining verifiable by a court or auditor years later. These are fundamentally different from blockchain workloads. The number of signers is typically in the tens or hundreds rather than the tens of thousands, and the signers may not use the same post-quantum scheme. An early adopter may already use ML-DSA, while another organization, following a different migration timeline, may use Falcon or a classical/post-quantum hybrid. FN-DSA, the NIST standard for Falcon, remains under development as FIPS 206 at the time of writing [29]. The signing process may also need to be fully non-interactive: each authority signs independently, using its own existing key, without requiring synchronization with the other signers. Finally, exact accountability is often essential. A court or auditor may need to determine which named authority signed a particular clause, rather than merely verify that at least t members of a predefined committee participated. These requirements raise three central questions: *Can existing aggregation techniques accommodate heterogeneous signers, can they preserve exact signer attribution without interactive coordination, and what communication, computation, and verification costs do they incur under realistic institutional workloads?*

Our survey finds that these questions are largely unanswered empirically. To our best knowledge, there exists no published evaluation that quantitatively studies the institutional workload across the aggregation paradigms considered in this paper. This does not establish that existing approaches are unsuitable; rather, it reveals a workload-evaluation gap that requires systematic investigation.

To address this gap, we present a Systematization of Knowledge (SoK) that organizes existing work around deployment requirements rather than cryptographic constructions alone. Our contributions are:

- **Taxonomy:** An eight-dimensional taxonomy covering interactivity, key model, scheme homogeneity, accountability, output compatibility, security assumptions, scale, and deployment constraints.
- **Systematization:** A systematic classification of more than 30 schemes published between 2021 and 2026, covering threshold and distributed signing,

proof-based aggregation, STARK-based aggregation, algebraic half aggregation, and related approaches.

- **Workload analysis:** Three falsifiable hypotheses that characterize the gap between blockchain-validator and institutional-signing workloads, including the lack of evaluations involving heterogeneous post-quantum schemes.
- **Benchmark methodology:** A common evaluation methodology covering communication, computation, verification, scalability, heterogeneity, and accountability across representative workloads.
- **Research agenda:** Concrete requirements and open problems for determining whether existing aggregation techniques can support heterogeneous, non-interactive, and accountable institutional signing.

Our goal is not to propose another aggregation construction, but to establish a common basis for answering a more fundamental question: *which post-quantum multi-party signing techniques actually match the requirements of the workloads in which they are intended to be deployed?*

The remainder of the paper is organized as follows. Section 2 provides the necessary background. Section 3 presents our taxonomy and review methodology, followed by the systematization in Section 4. Section 5 develops the workload-mismatch hypotheses, Section 6 examines security assumptions, and Section 7 presents the benchmark methodology. Sections 8–11 discuss the motivating workload, implications, future directions, and limitations, followed by the conclusion in Section 12.

2 Background

Compact multi-party signing in the post-quantum setting sits at the intersection of three areas: post-quantum signature schemes, multi-party signing models, and techniques for compressing the resulting signatures or their verification. Understanding their differences is important for interpreting the literature surveyed in this paper. In particular, the underlying signature scheme determines the structure of the verification relation, while the signing model determines whether signers operate independently or as participants in a coordinated protocol. Aggregation and succinct-proof techniques then provide different ways of reducing the cost of representing or verifying many signatures. This section introduces these concepts and establishes the terminology used throughout the remainder of the paper.

The three signature standards considered in this work have substantially different structures. **ML-DSA** (FIPS 204, finalized in August 2024 [27]) is a module-lattice-based signature scheme built on the Fiat–Shamir-with-Aborts paradigm and the Module-LWE and Module-SIS assumptions. During signing, the signer generates a masked commitment, derives a challenge using SHAKE-256, and computes a response. The signer rejects and retries when the response does not satisfy the required bounds, preventing information about the secret key from leaking through the transcript. Verification checks the resulting linear

relation together with norm bounds and auxiliary hint information. ML-DSA-65 has a 1,952-byte public key and a 3,309-byte signature.

FN-DSA, the standardized name for Falcon, follows a different hash-and-sign construction based on NTRU lattices. FIPS 206 remains in draft form at the time of writing [29]. Its key-generation procedure creates an NTRU trapdoor, which is subsequently used to sample a short vector from a discrete Gaussian distribution during signing. Falcon-512 has an 897-byte public key and an average signature size of approximately 666 bytes, substantially smaller than an ML-DSA-65 signature. The current FIPS 206 draft also differs from the pre-standardization Falcon specification in parts of the signing transcript, including salt sampling and public-key hashing. This distinction matters because several aggregation schemes were developed using the earlier Falcon specification; the implications are discussed further in Section 11.

SLH-DSA (FIPS 205, finalized in August 2024 [28]) takes a fundamentally different approach and relies only on hash functions. It combines one-time WOTS+ signatures and few-time FORS signatures within a Merkle hypertree. Its keys are small, typically 32–64 bytes, but its signatures are much larger, ranging from roughly 8 to 50 KB depending on the parameter set. Verification is dominated by hash evaluations. Thus, the verification structure of SLH-DSA differs considerably from those of the two lattice-based standards, and this difference directly affects the types of aggregation and proof techniques that can be used.

The distinction between the underlying signature schemes is only part of the problem. Multi-party signing can itself be realized in several different ways. An *aggregate signature* combines independently generated signatures (pk_i, m_i, σ_i) into a compact object σ_{agg} that can be verified using the corresponding public keys and messages. The messages and keys may be different for different signers. A *multi-signature* is the special case in which all signers authenticate the same message. In both cases, the individual signatures can be produced independently and aggregation can be performed afterwards, including by a party that did not participate in signing. This property is particularly relevant to non-interactive institutional signing.

A *threshold signature* uses a different model. A single logical signing key is distributed among multiple parties, typically through a distributed key-generation protocol, and a subset of at least t parties subsequently executes a signing protocol to produce one signature under the shared public key. Threshold signing therefore differs from aggregation in both its key model and its interaction requirements. Aggregation combines signatures produced under independently managed keys, whereas threshold signing operates one shared key through a coordinated protocol. This distinction is central to the taxonomy developed in Section 3.

A further class of approaches does not combine signatures directly. Instead, it produces a succinct proof that a collection of signature-verification relations has been satisfied. A succinct non-interactive argument allows a prover to demonstrate that a witness w satisfies a relation $\mathcal{R}(x, w)$ using a proof that is much

smaller than the underlying computation. The verifier can then check the proof without independently evaluating every instance. Lattice-native systems such as LaBRADOR [2] represent suitable relations using lattice-based commitments and proof techniques, making them well suited to lattice-based verification relations. Hash-based systems such as STARKs use a different construction based on algebraic representations, error-correcting codes, and hash-based commitments such as FRI. The choice of representation therefore affects how naturally a given signature scheme can be incorporated into a succinct proof system.

Folding schemes provide another mechanism for reducing the cost of proving many instances. Rather than proving each verification relation separately, folding combines multiple instances incrementally into an accumulated instance. Repeating this process allows a large collection of relations to be represented by a final proof whose cost grows much more slowly than the number of original instances. Recent lattice-based folding constructions consequently provide important building blocks for proof-based signature aggregation. These different approaches—direct aggregation, threshold signing, succinct proofs, and folding—operate under different assumptions about keys, interaction, signer coordination, and verification. These differences form the basis for the systematic comparison developed in the following sections.

3 A Taxonomy for Systematizing Compact Signing

The literature on post-quantum multi-party signatures uses different constructions to address different deployment requirements. A threshold scheme, for example, may reduce the signing key to a single shared public key but require an interactive protocol, while an aggregation scheme may allow independent signing but retain one public key per signer. Similarly, some proof-based approaches can compress a large number of signatures, but may require a new verification procedure or impose restrictions on the underlying signature scheme. Comparing these approaches therefore requires more than looking at signature size or asymptotic complexity.

We organize the surveyed schemes using eight dimensions that capture the properties most relevant to deployment. The dimensions cover not only the cryptographic construction itself, but also interaction, key management, signer heterogeneity, accountability, verification compatibility, security assumptions, and the scale demonstrated in practice. Together, they provide a common basis for comparing schemes that were originally developed for different workloads.

3.1 Eight Dimensions

Each scheme is classified along the following eight dimensions.

1. **Compression paradigm:** Threshold or distributed signing; proof-based aggregation; algebraic half-aggregation; bespoke multi-signature.
2. **Interactivity:** Fully non-interactive; constant-round (e.g., MuSig2-style); full interactive threshold protocol.

3. **Key model:** Pre-shared, DKG-based single virtual key; independent long-term keys usable ad hoc; a fixed, pre-enumerated validator set.
4. **Scheme homogeneity:** Requires one identical scheme and parameter set for every signer, or tolerates a mix.
5. **Accountability granularity:** Exact identity of every contributor; a coarse bitfield over a known fixed set; none.
6. **Output compatibility:** Verifies under an unmodified standard verification algorithm, or requires a new, scheme-specific verifier.
7. **Post-quantum assumption family:** Lattice; hash-based; other.
8. **Demonstrated scale:** The value of N actually tested in the source paper, as distinct from what its asymptotics permit.

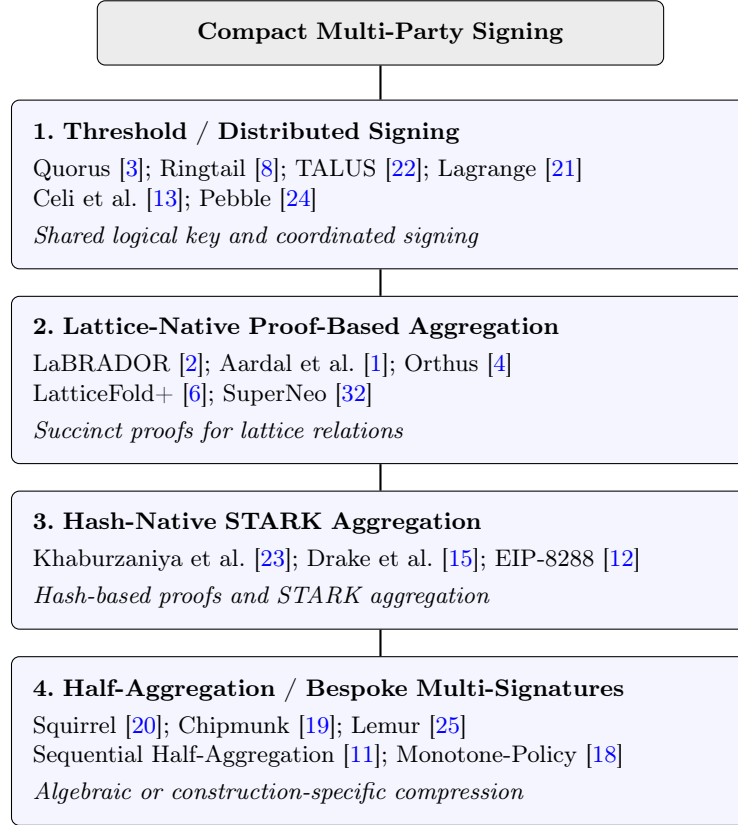


Fig. 1. Taxonomy of the four main paradigms for compact post-quantum multi-party signing. Representative constructions are shown for each paradigm; the complete classification is given in Section 4.

The eighth dimension is particularly important for this study because it allows the workload gap identified in Section 5 to be examined using reported experimental results rather than asymptotic claims alone. We therefore distinguish carefully between the scale supported in principle and the scale demonstrated experimentally. Figure 1 summarizes the relationship between the four main compression paradigms, while Figure 2 collects the reported values of N for schemes for which a directly comparable measurement is available.

3.2 Systematic Review Methodology

We surveyed the IACR Cryptology ePrint Archive and the proceedings of USENIX Security, IEEE S&P, ACM CCS, CRYPTO, EUROCRYPT, ASIACRYPT, PKC, and ESORICS for work published between 2021 and 2026. Searches combined terms including “threshold signature,” “aggregate signature,” “multi-signature,” “lattice,” “post-quantum,” “ML-DSA,” “Dilithium,” “Falcon,” and “hash-based.” The search period begins in 2021, which includes the first STARK-based hash-signature aggregation work considered in this survey [23].

We also followed references backward and forward from the identified works to capture related constructions and subsequent variants. Each candidate scheme was classified using the eight dimensions defined above. We recorded its construction, signing model, key assumptions, implementation details, and experimentally evaluated signer count where reported. Given the publication rate in this specific corner of the field over the last twelve months of the search window; three independent, mutually citing threshold-ML-DSA results appeared within a few months of one another, as Section 4 details this protocol as it is meant to be re-run immediately before submission, not treated as a one-time pass completed at the project’s start.

4 Systematization of Existing Approaches

The taxonomy in Section 3 provides the basis for organizing the existing approaches. We group the literature into threshold and distributed signing, lattice-native proof-based aggregation, hash-native STARK-based aggregation, algebraic half-aggregation, and bespoke multi-signatures. The following sections review the representative constructions in each branch and summarize their relevant properties, reported evaluations, and limitations with respect to the institutional workload mentioned in Section 2.

4.1 Threshold and Distributed Signing

Threshold ML-DSA has become an active sub-field within the last year, with recent work focusing on scale, round count, and setup assumptions. Quorus [3], from J.P. Morgan’s applied-cryptography group, is an MPC-based construction that outputs signatures verifying under the unmodified ML-DSA algorithm, providing strong compatibility with the standard. It is designed to scale to an arbitrary number of parties under an honest-majority assumption, with a concrete

demonstration around 64 parties. Its cost is dominated by 100 to 150 KB of per-party communication in each of the 37 to 136 rounds required by the underlying rejection-sampling protocol, resulting in a substantial aggregate communication cost. Celi, del Pino, Espitau, Niot, and Prest [13] consider the opposite point in this trade-off: their construction, to appear at USENIX Security 2026, is the first threshold scheme fully compatible with ML-DSA under a *dishonest*-majority assumption, at the cost of capping the party count at six, with average per-party communication of up to 1 MB.

Independently, Leo Kao, an individually credited researcher at Codebat Technologies in Taipei, published two solo-authored constructions addressing the same problem within a two-month period. The earlier masked Lagrange-reconstruction technique [21] targets arbitrary threshold T while producing standard, unmodified 3.3 KB ML-DSA signatures, explicitly addressing the round count of Quorus and the six-party limit of Celi et al. The later construction, TALUS [22], achieves one-round *online* signing by moving most of the computation to an offline phase, with better than 99% success in the first round. It also formalizes what its author calls the Lattice Threshold Trilemma, a proof that no group homomorphism from the ML-DSA nonce space into an abelian group can be simultaneously hiding and binding. This rules out an entire class of homomorphic-commitment approaches to the problem.

A cluster of related results fills out this branch: Trilithium [16] addresses the two-party case with a trusted randomness provider; del Pino and Niot’s PKC 2025 result [35] presents a compact lattice-based threshold signature; Ringtail [8] provides a two-round scheme directly from standard Module-LWE without Falcon- or Dilithium-specific machinery; Threshold Raccoon [33] and its identifiable-abort follow-up [34] identify the party responsible for a failed signing attempt; Flood-and-Submerge [17] provides a more general robust distributed key-generation framework; and Pebble [24] replaces Celi et al.’s hyperball-sampled partial signatures with discrete-Gaussian ones for a simpler entropy analysis. Every scheme in this branch requires interaction among the signers because one shared key is jointly operated rather than N independent keys.

4.2 Lattice-Native Proof-Based Aggregation

This branch keeps the N signers fully independent: each signs with an already-issued key, on an independent schedule, using an unmodified standard signing algorithm, while all compression is performed after signing. LaBRADOR [2] is the foundational succinct proof system in this line, achieving roughly 58 KB proofs for R1CS instances with up to 2^{20} constraints from the Module-SIS assumption. Aardal, Aranha, Boudgoust, Kolby, and Takahashi [1] instantiate this specifically for Falcon. They observe that Falcon’s verification relation does not embed naturally into R1CS, develop a “predicate special soundness” framework to provide a rigorous knowledge-soundness proof for the *non-interactive*, Fiat-Shamir-transformed version of LaBRADOR, and give the first complete aggregate-signature construction based on it.

Orthus [4] extends this approach further: a “Left-Right” protocol yields verification time that scales with the square root of the witness size rather than linearly. The paper reports a nine-fold verifier-time reduction over the Aardal et al. baseline at $N = 2^{17}$ aggregated Falcon signatures, with proofs of roughly 100 KB. Greyhound [30] provides fast lattice polynomial commitments as an alternative back end for the same class of relation. None of these evaluations tests a second signature scheme, a mixed-scheme signer set, or an N below the low hundreds. The smallest points on the Orthus scaling curve are still oriented toward a validator-scale setting, and Figure 2 shows this difference.

A parallel sub-branch provides general-purpose recursive machinery for this line of work. LatticeFold [5] introduced the first lattice-based folding scheme from Module-SIS; LatticeFold+ [6] (CRYPTO 2025) improved its prover performance by five to ten times while providing a simpler verifier and shorter proofs. Symphony [14], solo-authored by Binyi Chen at Stanford, is, to our knowledge, the first folding-based SNARK to avoid embedding a hash function inside the recursive circuit. Mangrove [26] further generalizes the folding-based SNARK framework, while Neo and SuperNeo [31,32] adapt HyperNova-style folding to small fields with pay-per-bit commitment costs and report satisfying six previously competing desiderata simultaneously. This sub-branch provides general proof-system components that can serve as back ends for aggregation rather than directly addressing the institutional signing workload.

4.3 Hash-Native, STARK-Based Aggregation

A structurally different approach applies to hash-based signatures. Since their verification cost is dominated by hash evaluations, a hash-native proof system, such as a STARK based on error-correcting codes and the FRI protocol, represents the verification relation naturally. Khaburzaniya, Chalkias, Lewi, and Malvai established this approach in 2021 [23]: an aggregator uses a STARK to prove the validity of many one-time hash-based signatures, producing a proof of roughly 100 to 200 KB regardless of whether thousands of signatures are aggregated. The authors also note that simple hash-based one-time-signature variants are particularly suitable for this approach.

This approach has also been considered for deployment. Drake, Khovratovich, Kudinov, and Wagner’s 2025 proposal for hash-based multi-signatures for post-quantum Ethereum [15] applies the same idea to validator-set deployment. The “hash-sig-agg” engineering effort, based on the Plonky3, SP1, and OpenVM STARK toolchains, is benchmarking this construction, while Ethereum’s EIP-8288 [12] proposes integrating it into the protocol as a post-quantum successor to BLS-based validator-signature aggregation. The surveyed works in this branch target validator-consensus workloads explicitly; none evaluates a mixed-scheme signer set or reports a concrete institutional-scale benchmark.

4.4 Algebraic Half-Aggregation

A fourth branch avoids a general-purpose proof system and combines signature components algebraically. Boudgoust and Roux-Langlois’s non-interactive half-aggregation [9] directly combines Fiat-Shamir-with-Aborts signature responses using a random linear combination. Their follow-up, “Overfull: Too Large Aggregate Signatures Based on Lattices” [10], documents where naive versions of this approach increase aggregate signature size substantially. Boudgoust and Takahashi’s sequential half-aggregation [11] extends the technique to both Dilithium-style and Falcon-style signatures. This branch does not require the succinct-proof machinery used in Sections 4.2 and 4.3, which simplifies implementation, but it provides weaker compactness than the proof-based approaches and requires careful control of aggregate size.

4.5 Bespoke Multi-Signatures and General Frameworks

A final branch designs new signing algorithms specifically for compactness, sacrificing compatibility with an existing standard verifier in exchange for other guarantees. Squirrel [20], from Fleischhacker, Simkin, and Zhang, introduced an efficient synchronized lattice multi-signature. Chipmunk [19], by Fleischhacker, Herold, Simkin, and Zhang, improved its aggregate size by a factor of 5.6, provides security against rogue-key attacks, supports independently generated keys, and compresses 8,192 signatures to roughly 136 KB under the Short Integer Solution assumption. Very recently, Lemur [25], from a Monash/Tampere collaboration, follows the Squirrel and Chipmunk approach while redesigning its foundations to improve efficiency, scalability, and flexibility. All three require synchronization, so signatures from different time steps do not aggregate together, and use signing algorithms that are not standard ML-DSA or Falcon.

Boneh and Kim’s interactive and one-time aggregate signature constructions [7] achieve aggregate sizes logarithmic in the number of signers, at the cost of either interactivity or a one-time-key restriction. At the more theoretical end, Farber Brodsky, Choudhuri, Jain, and Paneth’s Monotone-Policy Aggregate Signatures [18] show, from standard assumptions, that succinct aggregation is possible across signers using different preferred schemes and arbitrary monotone access policies. This establishes that heterogeneous-scheme aggregation is not fundamentally impossible. The construction is, however, an asymptotic plain-model result, with no concrete post-quantum instantiation and no benchmark at a specific value of N .

4.6 Cross-Cutting Synthesis

Table 1 summarizes the properties discussed in Sections 4.1–4.5. No row matches the institutional target on every column simultaneously. Monotone-Policy Aggregate Signatures matches the heterogeneity and accountability requirements but has not been instantiated concretely or evaluated at a specific scale. Figure 2 shows the same gap along a single axis, demonstrated scale, for the subset of schemes with a directly reported value of N .

Table 1. The taxonomy applied to representative schemes from each branch. “Institutional target” is the profile motivated in Section 1; no cataloged scheme matches it on every column. Demonstrated N is what the source paper actually measured, not its stated asymptotic reach.

Scheme	Paradigm	Interactive?	Heterog.?	Account.	Demo. N
Quorus [3]	Threshold (MPC)	Yes (many rounds)	No	Coarse	~ 64
Celi et al. [13]	Threshold	Yes	No	Coarse	~ 6
Kao, TALUS [22]	Threshold	Yes (1 online rd.)	No	Coarse	Small-scale demo
Ringtail [8]	Threshold	Yes (2-round)	No	Coarse	Small-scale demo
Aardal et al. [1]	Proof-based agg.	No	No	Exact	Large (paper-scale)
Orthus [4]	Proof-based agg.	No	No	Exact	2^{17}
Khaburzaniya et al. [23]	Proof-based agg. (hash)	No	No	Exact	Thousands
Boudgoust–Roux–Langlois [9]	Half-aggregation	No	Partial	Exact	Moderate
Chipmunk [19]	Bespoke multi-sig	No (synchronized)	No	Exact	8,192
Monotone-Policy [18]	General framework	No	Yes	Exact	Asymptotic only
Institutional target	—	No	Yes	Exact	5–200

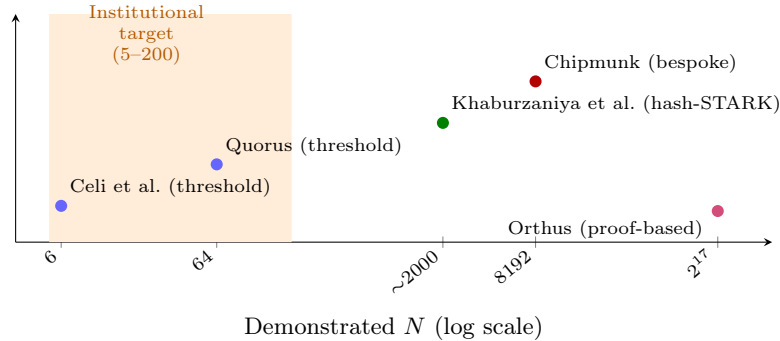


Fig. 2. Demonstrated scale N for schemes with a directly reported figure (Table 1), on a log axis. The two smallest points are also the only interactive (threshold) ones, so the paradigm reaching institutional scale is the one Section ?? rules out by requirement. Every non-interactive point sits an order of magnitude or more above the institutional band.

5 The Workload Mismatch

The systematization in Section 4 shows that existing approaches address different signing workloads and make different assumptions about signer scale, scheme homogeneity, interaction, and accountability. This section compares these assumptions with the institutional workload defined earlier and formulates three hypotheses that can be tested using a common benchmark.

5.1 Two Workloads, One Literature

Table 1 and Figure 2 indicate a mismatch between the workloads studied in the literature and the requirements of institutional document signing. The distinction is not only one of scale. The taxonomy dimensions also correlate with workload in the existing literature. Large N evaluations generally use homogeneous schemes, coarse accountability, and validator-oriented workloads, while

there is no comparable evaluation of small N , heterogeneous schemes with exact accountability. Confirming this mismatch requires running comparable systems under both workloads on the same hardware and using the same evaluation methodology. No existing paper performs such a comparison across the approaches considered in Section 4.

5.2 Hypotheses

- H1 (no free lunch).** The paradigm minimizing verifier time and output size at $N \in \{10^3, 10^4, 10^5\}$ with a homogeneous scheme and single message does not also minimize these quantities at $N \in \{10, 50, 200\}$ with mixed schemes, a single message, and full accountability.
- H2 (evaluation gap).** No existing published evaluation reports concrete measurements for a heterogeneous scheme signer set at any N . Unlike H1, this is directly checkable against the catalogue in Section 4 without new implementation.
- H3 (hidden setup cost).** Threshold schemes headline online signing results exclude the one time distributed key generation cost. This cost may be non negligible relative to the total end to end cost at institutional scale, where the amortization that justifies excluding it at validator scale may not apply.

Of the three hypotheses, H2 can already be assessed from Section 4. Across the cataloged schemes, no published evaluation reports a mixed scheme measurement. H1 and H3 require the benchmark described in Section 7.

6 Security-Model Systematization

Performance comparisons are incomplete without a corresponding security comparison. A faster scheme based on a weaker security assumption is not directly comparable to a slower scheme based on a stronger assumption. Table 2 records, for a representative subset of schemes, whether the soundness or unforgeability argument reduces to a standard lattice or hash assumption or relies on a heuristic argument; whether the construction assumes an honest majority of participating signers, as in Quorus, or tolerates a fully malicious signer set; whether it provides an explicit rogue-key defense, such as message augmentation, proof-of-possession, or key-set binding; and whether it provides identifiable abort or only an uninformative abort.

Rogue-key resistance is particularly relevant for constructions using independently chosen keys. The Boudgoust–Roux–Langlois half-aggregation line provides a relevant historical example, where the original construction was subsequently corrected to address this issue. Identifiable abort is another relevant property for threshold constructions; Unmasking TRaccoon [34], for example, explicitly identifies the party responsible for a failed signing attempt.

Two entries in Table 2 require additional context. Aardal et al. do not provide a fully general knowledge-soundness proof for their construction. Instead,

Table 2. Security-model comparison, built alongside the performance comparison of Table 1 rather than after it.

Scheme	Reduction	Majority	Rogue-key defense	Ident. abort
Quorus [3]	Standard (lattice)	Honest	N/A (single key)	No
Celi et al. [13]	Standard (lattice)	Dishonest	N/A (single key)	No
Unmasking TRaccoon [34]	Standard (lattice)	Dishonest	N/A (single key)	Yes
Aardal et al. [1]	Standard (relaxed KS)	N/A	Out of scope	N/A
Chipmunk [19]	Standard (SIS)	N/A	Yes, native	N/A
Boudgoust–Roux–Langlois [9]	Standard (lattice)	N/A	Corrected post-hoc	N/A

they prove a relaxed notion known as “somewhere extractability” in the non-interactive LaBRADOR setting. This notion is sufficient for their construction but is weaker than a standard knowledge-soundness statement. The distinction is therefore relevant when comparing their security argument with constructions that provide a stronger extraction guarantee.

The Boudgoust–Roux–Langlois line also illustrates the importance of explicitly considering rogue-key attacks in independently keyed constructions. The original construction was subsequently corrected to address this issue. This provides a concrete example of a security property that must be considered when evaluating aggregation schemes in which signers use independently generated keys.

7 Methodology for Empirical Resolution

7.1 Representative System Selection

Not every cataloged scheme can be re-benchmarked. Several results from 2025–2026 are still pre-publication or do not have publicly available implementations at the time of writing. We therefore select at least one representative construction from each paradigm in Section 4 for which either public code is available or the specification is sufficiently detailed for faithful reimplementations. The current candidates are the LaZer library together with the Aardal et al. Falcon relation for lattice-native proof-based aggregation; the “hash-sig-agg” project based on Plonky3 for hash-native STARK aggregation; Boudgoust–Roux–Langlois half-aggregation, which can be reimplemented from the published description; and Chipmunk, whose original paper reports a complete implementation, subject to confirming code availability at the start of the evaluation.

Threshold schemes, including Quorus, Celi et al., Kao’s constructions, and Ringtail, are included as documented reference points even though their interactivity does not satisfy the institutional workload requirement defined in Section 1. Including approaches that do not satisfy a particular workload constraint is necessary to characterize the full design space and identify where each paradigm ceases to meet the required conditions.

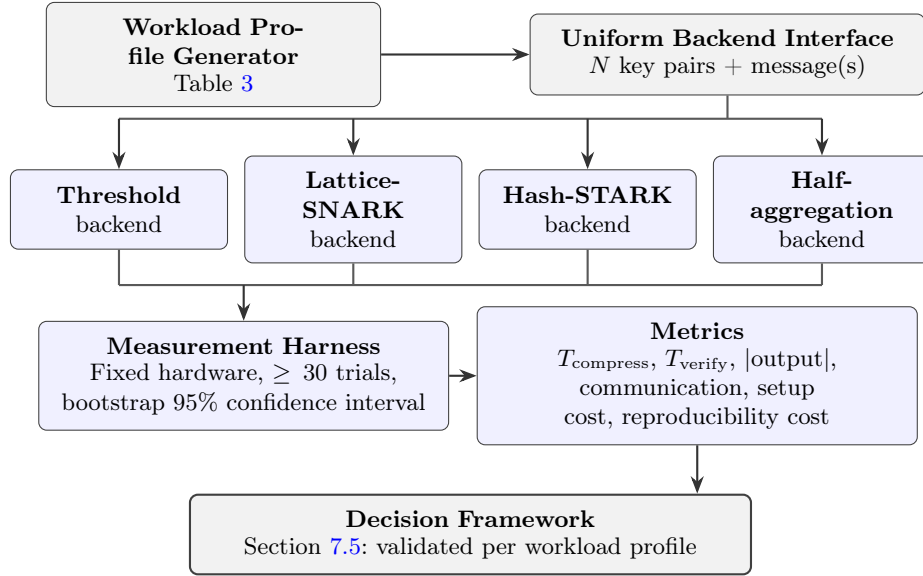


Fig. 3. Benchmark harness architecture. A workload profile is passed through a uniform backend interface to each aggregation paradigm. The resulting artifacts are evaluated using a common measurement harness and metric set, and the measurements are then used to validate the decision framework for each workload profile.

7.2 Benchmark Harness Design

Figure 3 shows the benchmark architecture. A common interface accepts N key pairs and a message, or message set, and each backend returns a compressed artifact together with timing and communication measurements. All backends are evaluated on the same hardware and under the same statistical protocol. This removes differences arising from hardware, trial counts, and measurement procedures that make direct comparison between independently reported results difficult.

7.3 Workload Profiles

Table 3 defines the three workload profiles. The example institutional mix of seventy percent ML-DSA-65, twenty percent Falcon-512, and ten percent transitional or hybrid schemes is not intended to represent measured migration statistics. Instead, it provides a concrete and reproducible workload for evaluating the heterogeneous migration scenario described in Section 1. The archival profile is included to evaluate long-term verification workloads involving SLH-DSA and other hash-based signatures. Its verification characteristics therefore represent a workload that is distinct from the validator-oriented evaluations discussed in Section 4.

Table 3. The three workload profiles used to test H1–H3. “Validator” represents the regime commonly evaluated in the existing literature; “Institutional” is the primary target; “Archival” emphasizes long-term re-verification.

Profile	N	Scheme mix	Message	Accountability
Validator	10^3 – 10^5	Homogeneous	Single, per round	Bitfield, fixed set
Institutional	5–200	Mixed (e.g. 70/20/10%)	Single document	Exact, all N
Archival	5–50	Mixed, incl. SLH-DSA	Single document	Exact, decades-scale

7.4 Metrics and Measurement Protocol

The primary metrics are T_{compress} (proving, signing, or aggregation time), T_{verify} , output size $|\text{output}|$, and communication cost where interaction is required. We additionally measure two properties that are not consistently reported in the existing literature. *Setup cost* denotes the one-time distributed-key-generation time and communication. It is zero by construction for the non-interactive paradigms considered in Sections 4.2–4.5. Reporting this cost separately avoids combining it with online signing measurements and allows H3 to be evaluated directly. *Reproducibility cost* denotes the engineering effort required to reproduce the reported results of a construction on the evaluation hardware within a predefined tolerance.

The statistical protocol uses fixed and fully reported hardware, including relevant instruction-set extensions such as AVX-512, since the LaZer LaBRADOR component requires these extensions and their absence can affect reproducibility. Each data point is evaluated using at least thirty trials and reported as a median with a bootstrap 95% confidence interval. Single-threaded and multi-threaded measurements are reported separately. Random seeds are fixed where the implementation permits this, and the benchmark harness and raw measurements are released as artifacts.

7.5 Deriving and Validating a Decision Framework

Using the combined performance results, extending Table 1, and the security properties summarized in Table 2, we derive a decision procedure based on the required range of N , scheme homogeneity, accountability, interactivity, and security-model requirements. The procedure maps a workload profile to the aggregation paradigms that satisfy its stated constraints and identifies the measured performance of those candidates.

The framework is then evaluated against the empirical results rather than treated as an independent assertion. For each workload profile in Table 3, we compare the framework’s selected candidate with the measured performance of all alternatives that satisfy the profile’s constraints. If the selected candidate does not provide the best measured result under those constraints, this is reported as a limitation of the framework and as an empirical finding. This validation separates conclusions supported by the benchmark from conclusions that follow only from the taxonomy or theoretical analysis.

8 Worked Example: Multi-Authority Document Signing

We apply the framework to the motivating scenario described in Section 1: a document requires signatures from N independent authorities, each holding a pre-existing and potentially different post-quantum key. The signers operate fully non-interactively, and the final result must provide exact accountability for all N signers. This workload corresponds to the institutional profile in Table 3, characterized by small N , heterogeneous schemes, non-interactive signing, and exact accountability.

The validation procedure in Section 7.5 determines which surveyed paradigms satisfy these requirements and how they compare on the measured performance and security metrics. If one or more existing paradigms satisfy all of the stated constraints, the benchmark identifies their performance and security characteristics for this workload. If no surveyed construction satisfies the complete set of requirements, the result provides empirical evidence that the institutional workload remains insufficiently covered by existing approaches.

In the latter case, the worked example provides a concrete formulation of the open problem rather than treating it as a general research gap. In particular, the missing combination of heterogeneous schemes, fully non-interactive signing, small signer sets, and exact accountability can be stated as a specific set of requirements for future aggregation constructions. The resulting observation is then connected to the open problems discussed in Section 10.

9 Discussion

The preceding sections establish the workload profiles, security properties, and empirical methodology needed to compare post-quantum multi-party signing approaches under common conditions. This section interprets the resulting comparison in the broader context of standardization and deployment. In particular, we discuss what the workload-based analysis implies for institutional adoption and why the current state of the post-quantum signature landscape makes such a comparison timely.

9.1 Implications for Standardization Bodies

NIST’s work on post-quantum threshold and multi-party signature functionality provides a broader standardization context for the constructions surveyed in Section 4. The surveyed literature, however, is largely evaluated using validator-oriented workloads. If H1 and H2 are supported by the empirical evaluation, this suggests that standardization guidance should distinguish between different deployment workloads when discussing threshold and aggregation mechanisms. In particular, institutional deployments such as banking, government, and legal document workflows may have requirements that differ from those of validator-consensus systems in terms of signer scale, scheme heterogeneity, interaction, and accountability.

Rather than referring to a threshold or aggregation scheme as a general solution, deployment guidance could therefore specify the workload assumptions under which a particular construction has been evaluated. This would allow institutional adopters to determine whether the underlying key model, interaction requirements, security assumptions, and accountability properties match their operational requirements.

9.2 Why Now: Recent Changes in the Design Space

The timing of this systematization is related to several developments in the post-quantum signature landscape during the period covered by our review. First, the standardization landscape has become sufficiently concrete to define specific deployment requirements. ML-DSA and SLH-DSA were finalized in August 2024, while FN-DSA remained in draft form during the period considered in this work. These developments provide a stable set of standardized or near-standardized schemes around which migration scenarios can be evaluated, including scenarios in which organizations adopt different schemes or parameter sets at different times.

Second, the threshold ML-DSA literature expanded substantially during the same period, with Quorus, Celi et al., and Kao’s constructions exploring different trade-offs in interaction, corruption assumptions, setup, and signing cost. This provides a sufficiently diverse set of constructions for a systematic comparison rather than a comparison centered on a single threshold design.

Third, the hash-based aggregation branch has progressed from the academic construction of Khaburzaniya et al. [23] toward protocol-level discussion, including Ethereum’s EIP-8288 [12]. This provides an additional example of how aggregation techniques developed for large validator sets are moving toward deployment-oriented evaluation.

Together, these developments make it possible to study post-quantum multi-party signing as a broader deployment problem rather than as a collection of isolated constructions. The resulting literature is sufficiently mature to support a systematic workload-based comparison, while the differences between validator and institutional requirements remain measurable through the methodology developed in Section 7.

10 Future Work and Open Questions

The analysis in this paper identifies several directions that can be pursued after the empirical evaluation. These directions fall into four categories: new constructions, broader workload evaluation, formal composition of existing techniques, and continued assessment of the security and deployment assumptions underlying the surveyed approaches.

10.1 The Conditional Construction Problem

If the empirical evaluation confirms that existing constructions do not support heterogeneous, fully non-interactive, individually accountable signing with independently managed keys, this result defines a concrete construction problem. The target would be a post-quantum signature-compression mechanism that supports heterogeneous underlying schemes, preserves individual signer accountability, requires no interaction among existing signers, and can be instantiated and evaluated at N in the tens to low hundreds. This would complement the asymptotic heterogeneous-scheme result of Monotone-Policy Aggregate Signatures [18] with a concrete construction and implementation.

One possible direction is to extend the proof-based aggregation approach of Section 4.2. Verification relations for a small set of post-quantum schemes, such as ML-DSA and Falcon, could be represented within a common proof framework and combined using a folding or recursive-proof back end. This would allow signers to retain independently managed keys while shifting the compression step to an aggregator. Another direction is to develop a policy-based construction closer to the framework of Monotone-Policy Aggregate Signatures, but with a concrete post-quantum instantiation and implementation. These approaches differ in both their proof structure and implementation requirements, making their comparative evaluation an open research question.

10.2 Beyond Two Workloads and Composition of Paradigms

The validator and institutional profiles considered in this paper represent two points in a broader workload space. Practical deployments may occupy intermediate regimes, such as permissioned consortium systems with hundreds of participants, heterogeneous validators, and stronger accountability requirements than conventional validator networks. Future evaluations can therefore parameterize the benchmark continuously in N , scheme heterogeneity, accountability, and interaction requirements. This would allow the performance and scalability of different paradigms to be studied across a broader range of deployment conditions.

The four paradigms identified in Section 4 can also be studied as potentially composable components rather than only as alternatives. For example, algebraic half-aggregation could reduce the number of objects presented to a subsequent succinct-proof system. Such a composition could combine properties of both techniques, but its security cannot be inferred simply from the security of the individual components. Establishing soundness, unforgeability, and rogue-key resistance for composed constructions is therefore a separate formal problem. The security properties summarized in Table 2 provide a starting point for identifying the assumptions that such a composition would need to preserve.

10.3 Security Assumptions and Long-Term Evaluation

The security properties of the aggregation layer should be evaluated alongside those of the underlying post-quantum signature schemes. Several proof-system

components in the surveyed literature rely on lattice assumptions whose concrete security at particular parameter choices may require further cryptanalytic study. Future benchmark versions should therefore record the security assumptions and parameter choices of each back end explicitly and update this information as the underlying cryptanalysis develops.

This consideration is particularly important for a decision framework intended to remain useful beyond a single benchmark release. Performance measurements can remain unchanged while the security interpretation of an underlying assumption evolves. Recording these assumptions separately from performance therefore allows future versions of the framework to incorporate new cryptanalytic results without changing the basic workload methodology.

10.4 Standardization and Deployment Evaluation

The workload-based analysis also provides a basis for more detailed deployment studies. If the benchmark confirms a systematic difference between validator-oriented and institutional workloads, future work can examine how these requirements should be represented in standardization and deployment guidance. Relevant dimensions include signer scale, scheme migration, key management, accountability, interaction, verification compatibility, and long-term validation.

Such studies could evaluate whether existing standardized signature schemes and emerging aggregation mechanisms provide sufficient interoperability for institutional document workflows. They could also examine operational questions that are outside the scope of the present SoK, including certificate management, archival verification, key rotation, revocation, and migration between classical and post-quantum signatures.

11 Limitations

The empirical conclusions are bounded by the availability of public implementations and by the level of detail provided in published specifications. Several recent results from 2025–2026 may not provide sufficient implementation material for independent benchmarking at the time of evaluation. Such schemes remain part of the systematization in Section 4, but may be excluded from the empirical comparison in Section 7.

Hardware-specific measurements may also limit the generality of the results. In particular, implementations that depend on instruction-set extensions such as AVX-512 may exhibit different performance characteristics on ARM, embedded processors, or other deployment platforms. The benchmark therefore reports hardware and software configurations explicitly and should not be interpreted as a universal performance ranking across architectures.

The literature survey is also bounded by its search period and the availability of publications during that period. Because the post-quantum aggregation literature is developing rapidly, the search protocol in Section 3.2 should be repeated

immediately before submission to capture relevant publications and implementations released after the initial survey. Finally, bibliographic details for secondary references should be verified against the final versions of the corresponding publications before submission.

12 Conclusion

Post-quantum multi-party signature research has expanded across threshold signing, proof-based aggregation, hash-based aggregation, algebraic compression, and bespoke multi-signatures, but existing evaluations remain largely focused on validator-oriented workloads. This SoK systematizes these approaches using a workload-based taxonomy covering interaction, key management, scheme heterogeneity, accountability, verification compatibility, security assumptions, and demonstrated scale. We identify the distinct requirements of institutional document signing and formulate testable hypotheses concerning performance, heterogeneous schemes, and setup costs. We further propose a common benchmark methodology that evaluates computation, communication, output size, security properties, setup cost, and reproducibility under controlled workload profiles. The resulting framework provides a basis for determining where existing approaches apply and where their assumptions do not match institutional requirements. If the empirical evaluation confirms the identified gap, it yields a concrete open problem: compact, non-interactive, heterogeneous, individually accountable post-quantum multi-party signing with independently managed keys.

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